

1. **MS:** 7.3 (a, c, k, m, p), 7.6 (c, i, j, k, l)

2. **From Sydsaeter and Hammond**

9.3.7. Find:

(a) $\frac{d}{dt} \int_0^t x^2 dx$

(b) $\frac{d}{dt} \int_t^3 e^{-x^2} dx$

9.5.3. Find:

(a) $\int_1^4 \sqrt{t} \ln t dt$

(b) $\int_0^2 (x - 2)e^{-\frac{x}{2}} dx$

(c) $\int_0^3 (3 - x)e^x dx$

3. **Additional Question:**

Suppose that the demand curve is $P = 50 - 0.1Q$ and the supply curve is $P = 0.2Q + 20$. Find the equilibrium price and compute the consumer and producer surplus.